ORCA-NTN: Secure Self-Healing and Cyber Incident Coordination for NTN-Backed Networks

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1. Introduction

Non-terrestrial networks (NTN) now carry everyday services. Mobile towers, payment terminals, hospitals, utilities and emergency services increasingly reach the core network over NTN backhaul, above all where fibre does not reach. NTN hardware and software are advancing fast, from new terminals and direct-to-device links to software that joins satellites, gateways and terrestrial networks into one system. Cyber security has not kept pace: 3GPP leaves feeder-link and inter-satellite-link security unspecified [14], and national NTN guidance is largely advisory [16].

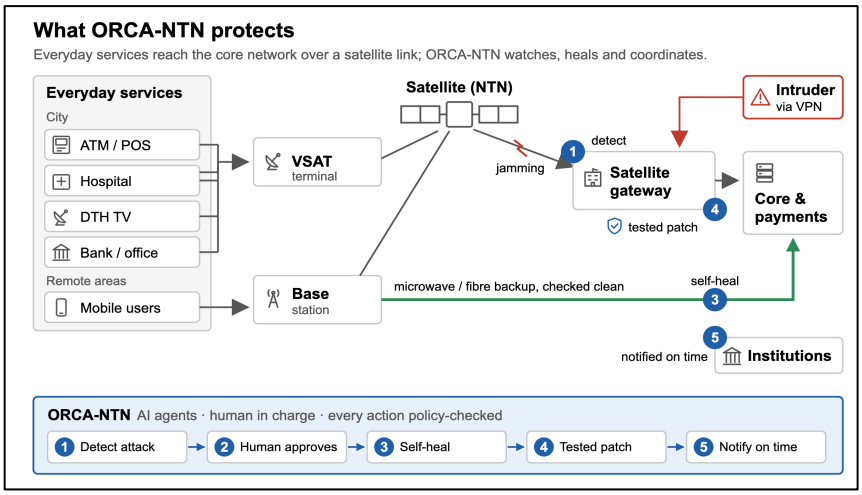


Figure 1. Everyday services over NTN and ORCA-NTN intervention points.

A fault and an attack look alike, yet trigger different legal duties. Fixed failover cannot tell weather from jamming, and may move traffic onto a backup path an attacker controls. Each operator, regulator and dependent service sees only part of the evidence, and the law is often unclear on who must act and by when. In the 2022 KA-SAT incident, an attacker used a misconfigured VPN to reach the operator's management network and disabled tens of thousands of terminals [1].

ORCA-NTN (Orchestrated Response and Coordination Agents) is a secure self-healing and cyber security incident-coordination system for NTN-backed networks. Its models detect and classify problems, one AI agent per institution reasons over the law, and a deterministic rule engine gives each legal verdict with its citation. A human approves every fix; the network heals only onto a path verified clean; and where the law is silent, ORCA-NTN records the gap with evidence and drafts a recommendation. In ten recorded scenarios it reached all 40 expected verdicts, with no fix run without approval. All links are emulated. India is the chosen country for the knowledge base, rules and gap register, with EU, US and UAE instruments as global comparisons.

2. Description of the use case

This section explains ORCA-NTN in detail. Figure 2 maps the workflow to ITU-T Y.3172 [2]; numbers refer to its steps. **Detect (1–4):** Emulated sites report link quality, rain, battery, logins, GNSS and packet loss each minute. These become 36 features, including an ITU-R P.838 rain-fade residual that helps distinguish weather from jamming [13]. M1–M4 analyse the evidence; alerts reach the relevant institutions. **Assess (5–7):** Institution agents retrieve KB evidence and call the deterministic YAML rule engine, which returns one of six verdicts with record citations. **Recover (8–11):** The owner proposes one of 17 playbooks. Human approval creates a one-use, exact-plan token valid for 15 minutes. Execution requires a clone test and clean-path check, followed by 5/25/100% rollout, re-attack checks and automatic rollback on failure. **Coordinate and review (12):** Agents draft notices and assemble evidence for human policy review. Humans identify gaps and sign recommendations. A hash-chained audit trail records every step.

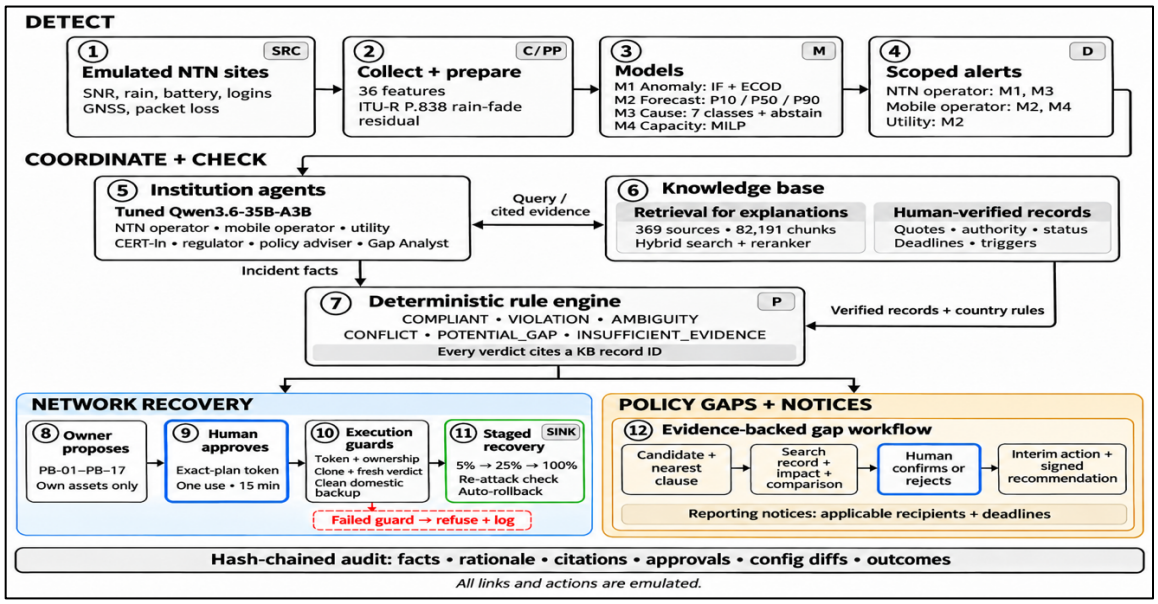


Figure 2. The ORCA-NTN pipeline.

Output. A dashboard shows affected sites, model alerts, agent explanations with cited policy checks, reporting deadlines, staged recovery and the audit trail.

Explore the ***ORCA-NTN Interactive Demo*** and watch the ***Complete 4K Video Walkthrough***.

ML models. The ML models continuously analyse NTN telemetry to detect anomalies, forecast network conditions and identify threats. M1 detects unseen attacks through unsupervised learning; M2 forecasts with uncertainty bounds; M3 identifies root causes using SHAP and defers uncertain cases to humans; and M4 applies verifiable deterministic optimisation. M1–M3 are trained and tested on NTN emulator data (Table 1), while M2 also uses ITU sandbox Turkcell data [3].

Table 1. The four models (held-out test set)

Model	Job	Method	Result
M1	Flags unusual behaviour at a site	Isolation Forest + ECOD	ECOD AUC 0.988; 2.3% false alarms
M2	Forecasts minutes left before the link fails	LightGBM P10/P50/P90 + conformal band	4.3 min error; 92.5% coverage
M3	Cause: weather, power/equipment, jamming, management plane, GNSS, SS7, or normal	Calibrated LightGBM + SHAP; abstains if unsure	97.1% accuracy; 1.7% abstain
M4	Shares scarce backup capacity by priority	MILP (HiGHS), greedy fallback	Emergency calls, hospital fully served

The agents. Each institution has a dedicated agent with defined powers (Table 2). Event-triggered agents analyse alerts, retrieve legal evidence, apply rules, propose playbooks, request approval and draft notices. Code validates tool calls and citations, while AI explains legal obligations, M3 findings and policy gaps. Humans resolve legal ambiguities, validate gaps and approve regulatory actions.

Hybrid ML and AI agents. The hybrid ML–LLM architecture uses task-specific ML models because they outperform general-purpose LLMs in numerical analysis, mathematical computations and pattern detection, while reducing compute and token costs. LLM agents interpret ML findings, retrieve legal evidence, explain decisions and recommend actions, governed by deterministic rules and human approval.

Table 2. What each agent sees, may do and may not do

Agent	Sees	May do	May not do
NTN operator	M1, M3 alerts; logins	Run its playbooks (PB-01–05, 11–14) once approved	Act without a token
Mobile operator	M2, M4 alerts	Failover PB-06, load shedding PB-15, warning PB-10, once approved	Use a foreign gateway
Utility, CERT-In, regulator	M2 alerts; shared facts	Check power risk, reporting and licence duties; draft notices	Run any fix
Policy adviser	Questions from agents	Answer from the knowledge base, with record ids	Answer without a source
Gap Analyst	Alerts, verdicts	Test legal coverage; record gaps with search record; draft recommendations	Confirm a gap
Commander (human)	Everything	Approve or reject each plan	Hand approval to an agent

Why Qwen. Agents use Qwen3.6-35B-A3B, QLoRA-tuned on our tool-use scenarios. Its Apache-2.0 licence enables in-country self-hosting. It delivers 83 tokens/s on one 24 GB L4 and achieved 93% tool-choice accuracy across 300 held-out cases, with no unsafe execution.

Knowledge base and RAG. Three layers: 369 official documents (India, EU, US, UAE, ITU, 3GPP) with authority, binding status and hash; human-verified records with exact quotes, deadlines and triggers, which alone feed the rule engine; and the gap register. Rules are one file per country, so a new jurisdiction needs no code. Retrieval covers 82,191 clause-level chunks, each prefixed with its law and clause path, using Qwen3-Embedding-8B vectors, keyword search and a reranker; it jumps to a named clause and says “not found” when nothing fits (recall@5 0.70; exact citations 5 of 5).

Why is it novel, innovative

NTN hardware and software advance fast; NTN cyber security is still being developed. ORCA-NTN fills that gap:

- **Fault vs. Attack**: AI distinguishes failures from attacks to trigger appropriate legal responses.
- **Live Policy Gap Detection**: Identifies regulatory gaps with supporting evidence.
- **Secure Self-Healing**: Verifies backup safety and data residency before failover.
- **Tested Automated Fixes**: Validates patches before deployment, with rollback support.
- **Law-Driven AI**: Rule engine makes legal decisions; LLM only explains and drafts.

3. Use case requirement

Technical Req.: Python 3.12 runs M1–M4 and the React dashboard. Offline mode requires no LLM; live AI uses Qwen. ChromaDB supports knowledge retrieval. **Functional Req**.: Detect anomalies, forecast failures, classify threats, optimize capacity and assess regulatory duties. Agent recommendations require citations, human approval, safety checks, rollback and audit logs.
Dataset Req.: Synthetic Chennai NTN telemetry, labelled attack scenarios, ITU Turkcell forecasting data and verified legal documents. Training and testing remain separate; no live operator data is needed. **Run**: `pip install -r requirements.txt` → `python -m orca.live --scenario S1 --llm mock --port 8060`. Open `http://127.0.0.1:8060`. See `docs/DEMO.md`.

4. Mapping with ITU-T Y.3172 Readiness 2.0 Report, and domain-specific policies

Standards mapping e.g. ITU-T Y.3172

Nine selected Readiness 2.0 dimensions [5] map to implemented capabilities: cross-domain evidence (D3), context (D4), human interface (D6), role objectives (D7), AI collaboration (D8), human decisions (D9), policy checks (D10), emergency-service protection (D11) and digital infrastructure (D13).

Table 3. Y.3172 nodes, components, AI Readiness 2.0 dimensions [5] and documents

Node	Use case (component and data)	Readiness 2.0	Documents
SRC	Emulated NTN sites, per-minute KPIs; Turkcell link data	Open data factor	CERT-In Direction (iv), logs [4]
C	One collector per institution; live event injection	D7 strategy; D4 context	IT Act s.70B(4), (6) [7]; CERT-In Directions (ii), (iii) [4]
PP	36 features, incl. rain-fade residual	D3 cross-domain	ITU-R P.838-3 [13]; DPDP Act [6]
M	M1 to M4; fine-tuned Qwen agents; RAG	D3; D13 infrastructure	3GPP TR 33.700-29 [14]
P	YAML rules, six cited verdicts, human approval	D10 policy; D6 human interface	CERT-In Dir. (ii) [4]; TCS Rules 7(1) [8]; Telecom Act s.20 [9]; NIS2 Art. 23 [10]
D	Owner routing; hash-chained audit trail	D8 collaboration; D9 humans	ITU-T Y.3172 [2]; IT Act s.70B(4) [7]
SINK	Failover, fixes, warnings, drafted notices	D11 inclusion	CTI Rules 7(1)(j) [11]; Authorisation Rules 60(1)(b) [15]; TDRA emergency calls [12]

5. Error scenarios

Scenarios for demo

Table 4. Ten reproducible scenarios; gaps refer to Table 5.

Scenario	Event and response	Question / gap
S0 Normal day	Healthy links; verify duties and domestic routing.	Negative control; covered duties, no invented incident.
S1 Terminals wiped	Detect management intrusion; sequence containment, failover and warnings.	clocks; warning; baseline.
S2 Storm / scarcity	Forecast rain and battery risk; M4 protects emergency and hospital capacity.	priority; battery.
S3 Unsafe backup	Identify jamming; reject unsafe/foreign routes; approve clean domestic path.	link standards; clean path.
N1 Battery cliff	Forecast hospital-corridor battery exhaustion; coordinate recovery and warnings.	battery; warning.
N2 Payments lost	Recover from a routing fault affecting payment terminals and ATMs.	warning.

Scenario	Event and response	Question / gap
N3 Feeder weakness	Audit an unprotected link despite healthy traffic; move to protected backup.	link standards; baseline.
N4 Dashboard lies	Cross-check false management metrics against independent traffic; contain and recover.	baseline; clean path; warning; action conflict.
N5 Several clocks	Track telecom/grid reporting triggers; use earliest applicable deadline.	clocks; baseline; warning.
CAP Double crisis	Combine rain, intrusion, battery loss and scarce capacity; coordinate recovery.	All seven confirmed gaps.

Tuned Qwen reached 40/40 expected verdicts with rules, versus 0/40 without, across ten scenarios. No playbook executed in either comparison arm. Judges can inject events live; approved recovery is demonstrated separately.

6. Policy gaps and standards gaps derived from the mapping exercise Standards and policy gaps

Agents retrieve relevant clauses and explain coverage limits; humans identify gaps and approve recommendations. Each is detected (POTENTIAL_GAP, AMBIGUITY or CONFLICT), proved with a search record, analysed for impact and EU comparison, acted on with an interim playbook, and turned into a recommendation a person signs. Table 5 lists the seven confirmed gaps for India (as of 7 Nov 2026).

Table 5. Policy and standards gap register, India

Gap	What is missing (evidence)	Interim action
Clocks	Up to four reports to three bodies, three triggers (noticing, awareness, occurrence) [4], [8], [11]	File at the earliest deadline
Warning	No duty for an NTN operator to warn dependent services; NIS2 Art. 23 has one [10]	PB-10 warning (Authz Rule 37(5) [15])
Priority	DoT SOP names priority callers, but no ranking for scarce backup capacity	PB-06, emergency calls first
Battery	“Sufficient” backup power, no minimum hours; grid nodes need 10 h (CEA)	PB-15: shed non-critical load
Links	Feeder and inter-satellite links “not specified” [14]; no s.19 standard [9]	Flag links with no security standard
Clean path	No rule requires a clean-path check before failover	PB-06 clean-path check
Baseline	NTN cyber framework is advisory [16]; no auditable baseline	Flag missing controls

Standards gaps and global comparison. No standard machine-readable format exists for policy rules, so ORCA-NTN defines its own YAML schema. NIS2 sets 24-hour and 72-hour reports [10]; the UAE TDRA gives emergency calls the highest priority at interconnection [12], and no UAE NTN-specific cyber rule was found. Every confirmed gap has global examples: 19 from the EU, US, California, UAE and ITU (kb/gaps/COVERAGE.md).

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